

01-iframe

iFrame

Iframe is two separate html code file integrated into one using `iframe` element.

Handling iFrame

There are two ways to handle iframe.

1. frameLocator
2. frame object
3. contentFrame

```
<html>
  <iframe class="outerFrame" name="outerFrameName">
    <iframe class="innerFrame" name="innerFrameName">
      <button id="btn">Click me</button>
    </iframe>
  </iframe>
</html>
```

frameLocator

- auto waits for elements
- reduces flakiness
- works well with nested frames
- meta data is not present using frameLocator

```
const frame = await page.frameLocator(".innerFrame");
frame.locator("#btn").click();
```

frame Object

Using frame object we get extra methods to operate on frames such as

- frame name
- Url
- childFrame
- parentFrame
- isDetached
- page

Any frame can be selected using two methods in frame object

- by url
- by name

By URL

```
const frame = page.frame({ url: "https://frame-url.com" });
frame.locator("#btn").click();
```

By Name

```
const frame = page.frame("innerFrameName");
frame.locator("#btn").click();
```

Different Methods Present In Frame

```
const frame = page.frame("frameName");
frame.url();
frame.name();
frame.childFrame();
frame.parentFrame();
frame.isDetached();
```

All Frames

To get all frames we have method `page.frames()`. This will get all the frames inside main frame or we can also provide `name` or `url` inside `frames()` method.

```
const frames = page.frames();
```

Frame traversing

frame traversing doen by

- `frame.paretnFrame()`
- `frame.page()`

where `frame.pratentFrame()` it traverse immediate frame one level up and `frame.page()` it switch all context to main page instead of one step at a time.

frameLocator is most preffered way to interact with iframe in playwright. But they dont provide meta inforamtion like name, url, isDetached for that we use frame object.

iFrame in selenium

```
// 1. By index (order in the DOM, 0-based)
driver.switchTo().frame(0);

// 2. By name or id attribute
driver.switchTo().frame("frameNameOrId");

// 3. By WebElement (most robust - locate the iframe like any element)
WebElement frame = driver.findElement(By.cssSelector("iframe#myFrame"));
driver.switchTo().frame(frame);
```

The WebElement approach is preferred because index is fragile (breaks if frame order changes) and not every frame has a name/id.

- `defaultContent()` — jumps all the way back to the top-level page.
- `parentFrame()` — moves up just one level (useful for nested frames).

Waiting for a frame (recommended)

Frames may load asynchronously, so combine with `WebDriverWait`:

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt(By.id("myFrame")));
```

Common gotchas

- `NoSuchElementException` when an element "should" be there → you're usually in the wrong frame context. Check whether you forgot to `switchTo().frame(...)` or forgot to `switchTo().defaultContent()`.
- After switching into a frame, all subsequent `findElement` calls are scoped to that frame until you switch out.

Comparison Tables

Playwright: `frameLocator` vs `frame` Object

Aspect	<code>frameLocator</code>	<code>frame</code> Object
Auto-waits for elements	Yes	No
Flakiness	Low (reduces flakiness)	Higher
Nested frames	Works well	Supported but manual
Frame metadata (name, url, etc.)	Not available	Available
Selecting a frame	By selector	By name or url

Aspect	frameLocator	frame Object
Extra methods (childFrame, parentFrame, isDetached, page)	No	Yes
Best for	Simple element interaction	Inspecting/traversing frames

Selenium vs Playwright iFrame Handling

Aspect	Selenium (Java)	Playwright
Model	Stateful — switch context in/out	Stateless — chain straight in
Switch into frame	<code>driver.switchTo().frame(.)</code>	<code>page.frameLocator(.) /</code> <code>page.frame(.)</code>
Switch back out	Required (<code>defaultContent()</code> / <code>parentFrame()</code>)	Not needed
Auto-wait	No (need <code>WebDriverWait</code>)	Yes (built-in)
Select frame by	Index, name/id, WebElement	Selector, name, url
Scope after switching	All lookups scoped to frame until switch out	Only chained locator is scoped
Common pitfall	Wrong frame context / forgot to switch out	Few — no context to manage

Playwright Methods — Quick Revision

Method	What it does
<code>page.frameLocator(selector)</code>	Returns a frame locator (auto-waits) to act on elements inside the frame
<code>page.frame(name)</code>	Get a frame object by its name
<code>page.frame({ url })</code>	Get a frame object by its url
<code>page.frames()</code>	Returns array of all frames on the page
<code>frame.locator(selector)</code>	Locate an element inside the frame
<code>frame.url()</code>	Get the frame's url
<code>frame.name()</code>	Get the frame's name
<code>frame.childFrame()</code>	Get the child (nested) frame
<code>frame.parentFrame()</code>	Move one level up to the parent frame

Method	What it does
<code>frame.isDetached()</code>	Check if the frame is detached from DOM
<code>frame.page()</code>	Jump straight back to the main page context

Playwright → Selenium (Java) Method Mapping

Selenium has no first-class frame object, so metadata is read via `getAttribute(. )` or the JavaScript executor.

Playwright	Selenium (Java) equivalent
<code>frame.url()</code>	<code>frameElement.getAttribute("src")</code> or JS <code>document.URL</code>
<code>frame.name()</code>	<code>frameElement.getAttribute("name")</code> or JS <code>window.name</code>
<code>frame.parentFrame()</code>	<code>driver.switchTo().parentFrame()</code>
<code>frame.page()</code>	<code>driver.switchTo().defaultContent()</code>
<code>frame.childFrame()</code>	switch into the nested frame (<code>switchTo().frame(.)</code>)
<code>frame.isDetached()</code>	no direct equivalent (catch <code>StaleElementReferenceException</code>)
<code>page.frames()</code>	<code>driver.findElements(By.tagName("iframe"))</code> (returns elements, not frame objects)
<code>page.frameLocator(sel).locator(.)</code>	<code>switchTo().frame(.)</code> then <code>findElement(.)</code>